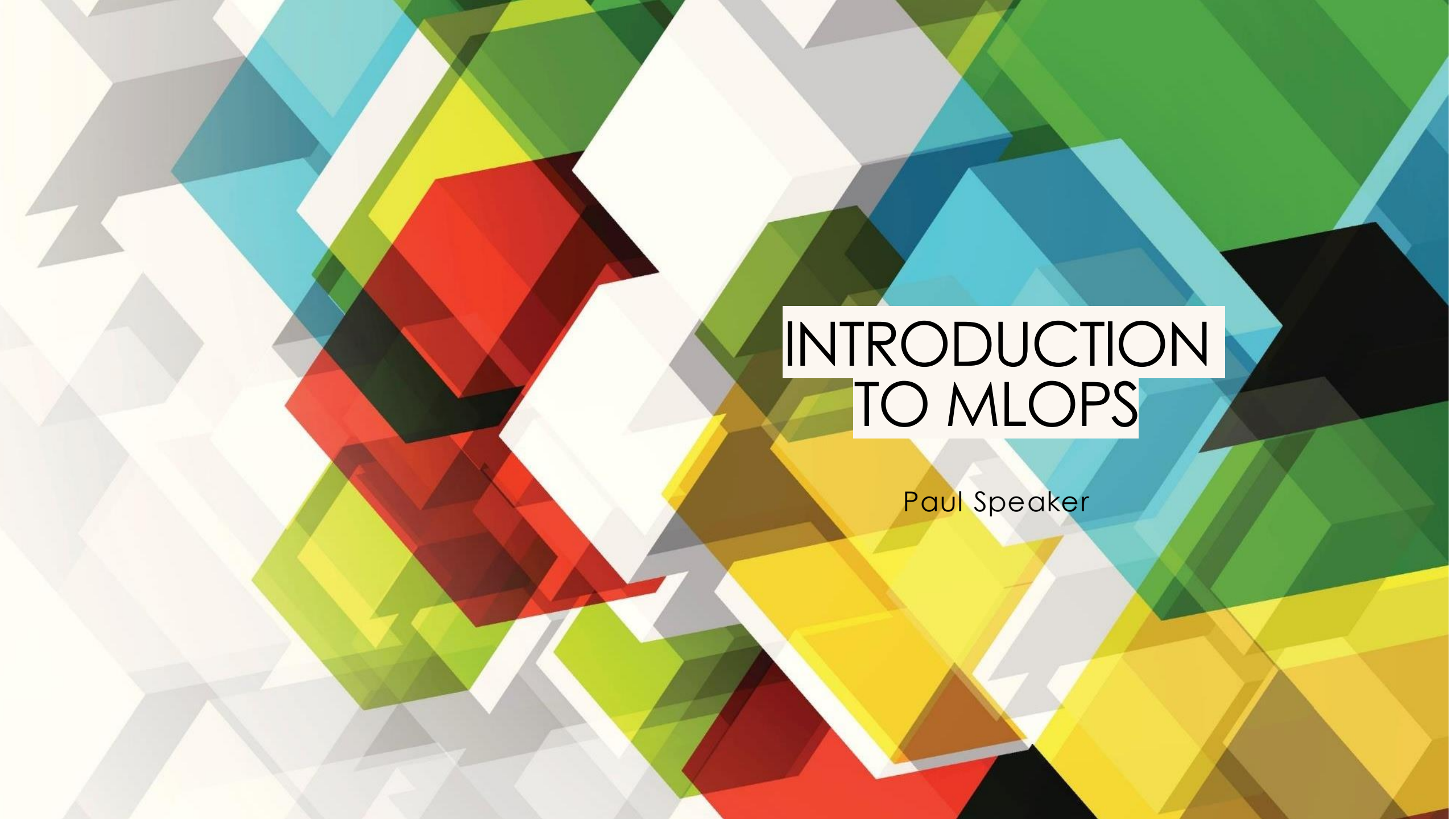




INTRODUCTION TO MLOPS

Paul Speaker

Welcome to MLOps!

- End-To-End process of creating a solution which includes a ML algorithm
- Creating a Data Science model is only part of that process!

Data Science Problem/Kaggle	MLOps
Problem is handed to you.	Figure out what the problem is.
Data is handed to you.	Figure out what data you need and how to get it.
Find best model according to accuracy metric	Find best model according to institutional constraints.
Turn in assignment—Done	Figure out how to get model in hands of users.
Done	Ensure that model is working as expected.
Done	Fix model when (not if) performance degrades.

Myth vs. Reality

This takes time!!! Took him 5 years to get his data!

Sellers sell data, customers will make fake orders to hold their place in line, may have thousands of orders in their name with out any shipment

- Myth: I'm going to build the model.
- Reality: I'm going to manipulate the data to build the model.
- Oh wait:
 - I'm going to get the data so that I can manipulate it to build the model
- But there's more: I'm going to get the permission to get the data to build the model
- And it continues: I'm going to ask the person from whom I can get the permission to get the data to build the model

MLOps as Logistics

- The best model in the world does no good if it cannot get to where it is needed when it is needed
 - Many MLOps problems are logistical problems in disguise
 - What could go wrong?
- “Where it is needed”—the end user
 - Can be a person who is using model to make decisions
 - Loan default model—decision on whether for bank to give person loan
 - Can be an automated system
 - Autonomous taxi
- There are a lot of steps that involve “movement”—logistics
 - Moving data, predictions, etc.
 - Have to be aware of timing rules
 - Also have to be aware of what can go wrong

The Big Questions for MLOps

Why: E.g., pizza delivery

if predict when pizza will arrive and
model says 7:30, better to be earlier
than later for customer satisfaction

Maybe take that into account with model
creation/selection, skew to be slightly early

- Why: what is the purpose for the model? The answer is never “to predict X!”
- Where: Where are the results needed?
- What: What form do the predictions take to be useful?
- When: When are the predictions needed (and how often?)
- Who: Who needs them (more important: who needs to like them?)
- How: the process of doing everything here
- Let's talk about these in the context of specific problems:
 - Customer loan default prediction
 - Demand forecasting
 - Recommender System

Some Important Terminology for MLOps

- Environments—where computing occurs
- Dev/Prod—twin terms developing to Development and Production.
Development is the creation of the model, and Production is getting model results out to those who are using it
- Productionization—putting the model predictions
 - where they needed,
 - when they are needed
 - To whom they are needed
 - With the accuracy that is needed
- Architecture—the high level description of the systems all needed to pull things together.

Do GenAI's Make MLOps Better?

MIT report: 95% of generative AI pilots at companies are failing



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Most Common Mistakes in MLOps

- “It’s somebody else’s problem.”
- Failing to Plan when you plan, you are anticipating obstacles
- Fake Internet Consensus Syndrome if you go to online forums, it always looks like people do MLOps work in the same way, but these are not representative of how data science works
- Modelophilia
- Outlier dismissal models may have great accuracy, but the few errors are the ones the client cared about the most
- Data Scientist-Centered Design you are not the user!!
- Vendor panaceas lots of people sell AI and ML solutions, 90% of them are liars and these “solutions” never work out
- LLM panaceas barrier to success with LLMs is consistency. they are not consistent enough. consistency >> accuracy in most cases